

$$\vec{L} = \vec{r} \times \vec{p}$$

$$[L^2, L_\alpha] = 0$$

$$\alpha = x, y, z$$

$$[L_\alpha, L_\beta] = i\hbar \epsilon_{\alpha\beta\gamma} L_\gamma$$

$$[L^2, L_z] = 0$$

$$m = -l, \dots, l$$

$$\rightarrow L^2 |l, m\rangle = \hbar^2 l(l+1) |l, m\rangle$$

$$L_z |l, m\rangle = \hbar m |l, m\rangle$$

$$|l, m\rangle \rightarrow Y_{lm}(\vartheta, \varphi) = P_l(\cos \vartheta) e^{im\varphi}$$

$$L_\pm = L_x \pm iL_y, \quad L_+ + L_-$$

$$L_\pm |l, m\rangle = \hbar \sqrt{l(l+1) - m(m \pm 1)} |l, m \pm 1\rangle$$

$$\textcircled{1} \quad l=1$$

$$H = -\vec{\mu} \cdot \vec{B} = -\lambda \vec{L} \cdot \vec{B}$$



$$(\vec{L} \cdot \hat{n}) |\psi_0\rangle = \hbar |\psi_0\rangle \text{ max. proj.}$$

$$H = -\lambda L_z B$$

$$\hat{n} = \begin{bmatrix} \sin \vartheta \\ 0 \\ \cos \vartheta \end{bmatrix}$$

$$|\psi_0\rangle = \alpha |1, -1\rangle + \beta |1, 0\rangle + \gamma |1, 1\rangle$$

$$\rightarrow H |1, m\rangle = -\lambda B \hbar m |1, m\rangle$$

$$|\psi, t\rangle = \alpha e^{-i\lambda B t} |1, -1\rangle + \beta e^0 |1, 0\rangle + \gamma e^{i\lambda B t} |1, 1\rangle$$

$$e^{-i\frac{E}{\hbar}t}$$

$$\text{ZP: } \vec{L} \cdot \hat{n} = L_x \sin \vartheta + L_z \cos \vartheta$$

$$(\vec{L} \cdot \hat{n}) |\psi_0\rangle = (L_x \sin \vartheta + L_z \cos \vartheta) (\alpha |1, -1\rangle + \beta |1, 0\rangle + \gamma |1, 1\rangle) = \hbar (\alpha |1, -1\rangle + \beta |1, 0\rangle + \gamma |1, 1\rangle)$$

$$L_x = \frac{1}{2} (L_+ + L_-)$$

$$L_+ |1, -1\rangle = \sqrt{2} \hbar |1, 0\rangle$$

$$L_- |1, -1\rangle = 0$$

$$\begin{aligned} L_z |1, -1\rangle &= -\hbar |1, -1\rangle \\ L_z |1, 0\rangle &= 0 \\ L_z |1, 1\rangle &= \hbar |1, 1\rangle \end{aligned}$$

$$L_+ |1, 0\rangle = \sqrt{2} \hbar |1, 1\rangle$$

$$L_- |1, 0\rangle = \sqrt{2} \hbar |1, -1\rangle$$

$$L_+ |1, 1\rangle = 0$$

$$L_- |1, 1\rangle = \sqrt{2} \hbar |1, 0\rangle$$

$$\begin{aligned} L_x |1, -1\rangle &= \frac{\sqrt{2}}{2} \hbar |1, 0\rangle \\ L_x |1, 0\rangle &= \frac{\sqrt{2}}{2} \hbar (|1, 1\rangle + |1, -1\rangle) \\ L_x |1, 1\rangle &= \frac{\sqrt{2}}{2} \hbar |1, 0\rangle \end{aligned}$$

$$\rightarrow \alpha \frac{\sqrt{2}}{2} \hbar \sin \vartheta |1, 0\rangle + \beta \frac{\sqrt{2}}{2} \hbar \sin \vartheta (|1, 1\rangle + |1, -1\rangle) + \gamma \frac{\sqrt{2}}{2} \hbar \sin \vartheta |1, 0\rangle - \alpha \hbar \cos \vartheta |1, -1\rangle + \gamma \hbar \cos \vartheta |1, 1\rangle$$

$$|1, -1\rangle: \beta \frac{\sqrt{2}}{2} \hbar \sin \vartheta - \alpha \hbar \cos \vartheta = \alpha \hbar \rightarrow \alpha = \beta \frac{\sin \vartheta}{1 + \cos \vartheta} \frac{\sqrt{2}}{2} = \beta \frac{\sqrt{2}}{2} \frac{\sin \frac{\vartheta}{2} \cos \frac{\vartheta}{2}}{2 \cos^2 \frac{\vartheta}{2}} = \beta \frac{\sqrt{2}}{2} \tan \frac{\vartheta}{2}$$

$$|1, 0\rangle: \alpha \frac{\sqrt{2}}{2} \hbar \sin \vartheta + \gamma \frac{\sqrt{2}}{2} \hbar \sin \vartheta = \beta \hbar$$

$$|1, 1\rangle: \beta \frac{\sqrt{2}}{2} \hbar \sin \vartheta + \gamma \hbar \cos \vartheta = \gamma \hbar \rightarrow \gamma = \beta \frac{\sin \vartheta}{1 + \cos \vartheta} \frac{\sqrt{2}}{2} = -\beta \frac{\sqrt{2}}{2} \cot \frac{\vartheta}{2}$$

$$\rightarrow |\psi_0\rangle = \beta \left(\frac{\sqrt{2}}{2} \tan \frac{\vartheta}{2} |1, -1\rangle + |1, 0\rangle - \frac{\sqrt{2}}{2} \cot \frac{\vartheta}{2} |1, 1\rangle \right)$$

$$1 = \langle \psi_0 | \psi_0 \rangle = |\beta|^2 \left(\frac{1}{2} \tan^2 \frac{\vartheta}{2} + 1 + \frac{1}{2} \cot^2 \frac{\vartheta}{2} \right) = |\beta|^2 \left(\frac{\sin^2 \frac{\vartheta}{2} + 2 \sin^2 \frac{\vartheta}{2} \cos^2 \frac{\vartheta}{2} + \cos^2 \frac{\vartheta}{2}}{2 \sin^2 \frac{\vartheta}{2} \cos^2 \frac{\vartheta}{2}} \right) = \frac{|\beta|^2}{2 \sin^2 \frac{\vartheta}{2} \cos^2 \frac{\vartheta}{2}}$$

$$\rightarrow \beta = \sqrt{2} \sin \frac{\vartheta}{2} \cos \frac{\vartheta}{2}$$

$$|\psi_0\rangle = \sqrt{2} \sin \frac{\vartheta}{2} \cos \frac{\vartheta}{2} \left(\frac{\sqrt{2}}{2} \tan \frac{\vartheta}{2} |1, -1\rangle + |1, 0\rangle - \frac{\sqrt{2}}{2} \cot \frac{\vartheta}{2} |1, 1\rangle \right) =$$

$$= \sin^2 \frac{\vartheta}{2} |1, -1\rangle + \sqrt{2} \sin \frac{\vartheta}{2} \cos \frac{\vartheta}{2} |1, 0\rangle + \cos^2 \frac{\vartheta}{2} |1, 1\rangle$$

$$|\psi, t\rangle = \sin^2 \frac{\vartheta}{2} e^{-i\lambda B t} |1, -1\rangle + \sqrt{2} \sin \frac{\vartheta}{2} \cos \frac{\vartheta}{2} |1, 0\rangle + \cos^2 \frac{\vartheta}{2} e^{i\lambda B t} |1, 1\rangle$$

$$L_z |l, m\rangle = \hbar m |l, m\rangle$$

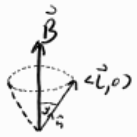
$$\rightarrow \langle L_z, t \rangle = \langle \psi, t | L_z | \psi, t \rangle = \langle \psi, t | \left(-\hbar \sin^2 \frac{\vartheta}{2} e^{-i\lambda B t} |1, -1\rangle + \hbar \cos^2 \frac{\vartheta}{2} e^{i\lambda B t} |1, 1\rangle \right) =$$

$$= \sin^2 \frac{\vartheta}{2} e^{i\lambda B t} \langle 1, -1 | (-\hbar) \sin^2 \frac{\vartheta}{2} e^{-i\lambda B t} |1, -1\rangle + \cos^2 \frac{\vartheta}{2} e^{-i\lambda B t} \langle 1, 1 | \hbar \cos^2 \frac{\vartheta}{2} e^{i\lambda B t} |1, 1\rangle = \hbar \left(\cos^2 \frac{\vartheta}{2} - \sin^2 \frac{\vartheta}{2} \right) = \hbar \cos \vartheta$$

$$\langle L_x, t \rangle = \langle \psi, L_x \psi \rangle = \dots = \text{Re} \langle L_x, t \rangle$$

$$\langle L_y, t \rangle = \dots = \text{Im} \langle L_y, t \rangle$$

$$\begin{aligned} \langle L_z, t \rangle &= \langle \psi, L_z \psi \rangle = \langle \sqrt{2} \sin \frac{\theta}{2} e^{-i\lambda B t} \langle 10 \rangle + 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \langle 11 \rangle \rangle = \\ &= 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} e^{-i\lambda B t} + 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} e^{-i\lambda B t} = \\ &= 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} e^{-i\lambda B t} = \hbar \sin \theta e^{-i\lambda B t} \end{aligned}$$



$$\begin{aligned} \langle L_x, t \rangle &= \hbar \sin \theta \cos \lambda B t \\ \langle L_y, t \rangle &= -\hbar \sin \theta \sin \lambda B t \end{aligned}$$

2D: $H = \frac{p^2}{2m} + \lambda(p_x s_y - p_y s_x)$, $S = \frac{\hbar}{2}$ Rashba Hamiltonian

$$\begin{aligned} p^2 &= p_x^2 + p_y^2 \\ H|\psi\rangle &= E|\psi\rangle \\ |\uparrow\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, |\downarrow\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \\ S_z |\uparrow\rangle &= \frac{\hbar}{2} |\uparrow\rangle, S_z |\downarrow\rangle = -\frac{\hbar}{2} |\downarrow\rangle \\ S_x |\uparrow\rangle &= \frac{\hbar}{2} |\downarrow\rangle, S_x |\downarrow\rangle = \frac{\hbar}{2} |\uparrow\rangle \\ S_y |\uparrow\rangle &= \frac{\hbar}{2i} |\downarrow\rangle, S_y |\downarrow\rangle = -\frac{\hbar}{2i} |\uparrow\rangle \end{aligned}$$

$$\begin{aligned} |\psi\rangle &= |\psi_\uparrow\rangle |\uparrow\rangle + |\psi_\downarrow\rangle |\downarrow\rangle \\ (H|\psi\rangle) &= E(|\psi_\uparrow\rangle |\uparrow\rangle + |\psi_\downarrow\rangle |\downarrow\rangle) \\ \frac{p^2}{2m} (|\psi_\uparrow\rangle |\uparrow\rangle + |\psi_\downarrow\rangle |\downarrow\rangle) &+ \lambda (p_x S_y - p_y S_x) (|\psi_\uparrow\rangle |\uparrow\rangle + |\psi_\downarrow\rangle |\downarrow\rangle) = E (|\psi_\uparrow\rangle |\uparrow\rangle + |\psi_\downarrow\rangle |\downarrow\rangle) \\ \frac{p^2}{2m} |\psi_\uparrow\rangle |\uparrow\rangle + \lambda (p_x \frac{\hbar}{2i} |\psi_\uparrow\rangle |\downarrow\rangle - p_y \frac{\hbar}{2} |\psi_\uparrow\rangle |\uparrow\rangle) &+ \frac{p^2}{2m} |\psi_\downarrow\rangle |\downarrow\rangle + \lambda (p_x \frac{\hbar}{2} |\psi_\downarrow\rangle |\uparrow\rangle - p_y \frac{\hbar}{2i} |\psi_\downarrow\rangle |\downarrow\rangle) = E (|\psi_\uparrow\rangle |\uparrow\rangle + |\psi_\downarrow\rangle |\downarrow\rangle) \\ \frac{p^2}{2m} |\psi_\uparrow\rangle + \lambda (p_x \frac{\hbar}{2i} |\psi_\downarrow\rangle - p_y \frac{\hbar}{2} |\psi_\uparrow\rangle) &= E |\psi_\uparrow\rangle \rightarrow -\frac{\hbar^2}{2m} \nabla^2 \psi_\uparrow + \lambda (-i \frac{\hbar}{2} \frac{\partial \psi_\downarrow}{\partial x} + \hbar \frac{\partial \psi_\uparrow}{\partial y} \frac{\hbar}{2}) = E \psi_\uparrow \\ \frac{p^2}{2m} |\psi_\downarrow\rangle + \lambda (p_x \frac{\hbar}{2} |\psi_\uparrow\rangle - p_y \frac{\hbar}{2i} |\psi_\downarrow\rangle) &= E |\psi_\downarrow\rangle \rightarrow -\frac{\hbar^2}{2m} \nabla^2 \psi_\downarrow + \lambda (\hbar \frac{\partial \psi_\uparrow}{\partial x} - i \frac{\hbar}{2} \frac{\partial \psi_\downarrow}{\partial y}) = E \psi_\downarrow \end{aligned}$$

→ dobili smo sistem dveh dif. enačb, ampak težko rešljiv
 $[H, \vec{p}] \neq 0$ če preizkusimo enostnost, 3 transkucijske simetrije → lastne funkcije so enake kot za $\vec{p}$ (namirom)

$$[H, p_x] = [\frac{p^2}{2m} + \lambda(p_x s_y - p_y s_x), p_x] = \lambda(p_x s_y - p_y s_x, p_x) = \lambda(p_x s_y - p_y s_x) = 0$$

$[H, p_y] = 0$ podobno

$$\begin{aligned} \vec{p} e^{i\vec{k} \cdot \vec{r}} &= (-i\hbar \nabla) e^{i\vec{k} \cdot \vec{r}} = -i\hbar i\vec{k} e^{i\vec{k} \cdot \vec{r}} = \hbar \vec{k} e^{i\vec{k} \cdot \vec{r}} \\ p|\vec{k}\rangle &= \hbar \vec{k} |\vec{k}\rangle \end{aligned}$$

$$\Rightarrow (\frac{p^2}{2m} + \lambda(p_x s_y - p_y s_x))(|\vec{k}\rangle (\alpha|\uparrow\rangle + \beta|\downarrow\rangle)) = E(|\vec{k}\rangle (\alpha|\uparrow\rangle + \beta|\downarrow\rangle))$$

$$\frac{\hbar^2 k^2}{2m} (\alpha|\uparrow\rangle + \beta|\downarrow\rangle) + \lambda (\hbar k_x \frac{\hbar}{2i} \alpha |\downarrow\rangle + \hbar k_y \frac{\hbar}{2} \beta |\uparrow\rangle) + \hbar k_x \frac{\hbar}{2} \alpha |\downarrow\rangle + \hbar k_y \frac{\hbar}{2i} \beta |\uparrow\rangle = E (\alpha|\uparrow\rangle + \beta|\downarrow\rangle)$$

$$\begin{aligned} \frac{\hbar^2 k^2}{2m} \alpha + \frac{\lambda \hbar^2 k_y}{2i} \beta - \frac{\lambda \hbar^2 k_x}{2} \beta &= E \alpha \\ \frac{\hbar^2 k^2}{2m} \beta + \frac{\lambda \hbar^2 k_x}{2} \alpha - \frac{\lambda \hbar^2 k_y}{2i} \alpha &= E \beta \end{aligned} \Rightarrow \begin{bmatrix} \frac{\hbar^2 k^2}{2m} & \frac{\lambda \hbar^2}{2i} (k_x - i k_y) \\ \frac{\lambda \hbar^2}{2i} (k_x + i k_y) & \frac{\hbar^2 k^2}{2m} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = E \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

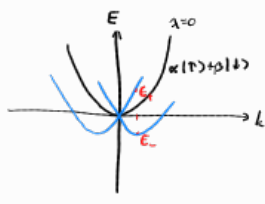
$$\underbrace{\begin{bmatrix} \frac{\hbar^2 k^2}{2m} & \frac{\lambda \hbar^2}{2i} k e^{-i\varphi} \\ \frac{\lambda \hbar^2}{2i} k e^{i\varphi} & \frac{\hbar^2 k^2}{2m} \end{bmatrix}}_{\text{hermitska}} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = E \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

Lastne vrednosti:

$$\begin{aligned} (\frac{\hbar^2 k^2}{2m} - E)^2 + (\frac{\lambda \hbar^2}{2i} k)^2 &= 0 \\ (\frac{\hbar^2 k^2}{2m} + \frac{\lambda \hbar^2}{2} k - E)(\frac{\hbar^2 k^2}{2m} - \frac{\lambda \hbar^2}{2} k - E) &= 0 \\ E_{\pm} &= \frac{\hbar^2 k^2}{2m} \pm \frac{\lambda \hbar^2}{2} k \end{aligned}$$

$$\begin{aligned} E_-: \begin{bmatrix} \frac{\hbar^2 k^2}{2m} - \frac{\lambda \hbar^2}{2i} k e^{i\varphi} & \frac{\lambda \hbar^2}{2i} k e^{-i\varphi} \\ \frac{\lambda \hbar^2}{2i} k e^{i\varphi} & \frac{\hbar^2 k^2}{2m} - \frac{\lambda \hbar^2}{2i} k e^{-i\varphi} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} &= 0 \\ \beta &= -\alpha e^{i\varphi} \\ \begin{bmatrix} \alpha \\ \beta \end{bmatrix} &= \alpha \begin{bmatrix} 1 \\ -e^{i\varphi} \end{bmatrix}, \alpha^2 + \beta^2 = 1 \Rightarrow \alpha = \frac{1}{\sqrt{2}} \\ |\psi\rangle &= \frac{1}{\sqrt{2}} (|\uparrow\rangle - e^{i\varphi} |\downarrow\rangle) \end{aligned}$$

$$E_+: \text{podobno} \Rightarrow |\psi\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle + e^{i\varphi} |\downarrow\rangle)$$



$$\begin{aligned} (S \cdot \hat{n})|\psi\rangle &= \frac{\hbar}{2} |\psi\rangle \\ \hat{n} &= (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta) \\ |\psi\rangle &= \cos \frac{\theta}{2} |\uparrow\rangle + \sin \frac{\theta}{2} e^{i\varphi} |\downarrow\rangle \end{aligned}$$

naš primer: $\theta = \frac{\pi}{2}$ dalec se giblje v xy ravnini

Lažji način (doluje za $S = \frac{\hbar}{2}$)

$$\begin{aligned} \vec{S} &= \frac{\hbar}{2} \vec{\sigma}, \vec{S} = (S_x, S_y, S_z) \\ \vec{\sigma} &= (\sigma_x, \sigma_y, \sigma_z) \\ &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \\ H &= \frac{p^2}{2m} + \lambda(p_x \frac{\hbar}{2} \sigma_y - p_y \frac{\hbar}{2} \sigma_x) \\ H(e^{i\vec{k} \cdot \vec{r}}) &= \frac{\hbar^2 k^2}{2m} e^{i\vec{k} \cdot \vec{r}} + \lambda (\frac{\hbar}{2} k_x \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} - \frac{\hbar}{2} k_y \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}) e^{i\vec{k} \cdot \vec{r}} \\ &= \begin{bmatrix} \frac{\hbar^2 k^2}{2m} & \frac{\lambda \hbar^2}{2} (-ik_x - k_y) \\ \frac{\lambda \hbar^2}{2} (ik_x - k_y) & \frac{\hbar^2 k^2}{2m} \end{bmatrix} e^{i\vec{k} \cdot \vec{r}} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \end{aligned}$$

Dodatek: jakien t jest P_{100} max?

$$\begin{aligned} \partial_t P_{100} &= 0 = \partial_t (c_1 t^2 e^{-c_2 t}) = c_1 t e^{-c_2 t} - c_1 t^2 c_2 e^{-c_2 t} \\ e^{-c_2 t} &= t e^{-c_2 t} c_2 \Rightarrow t = \frac{1}{c_2} = \frac{1}{\underline{\underline{c_2 - c_1}}} \end{aligned}$$